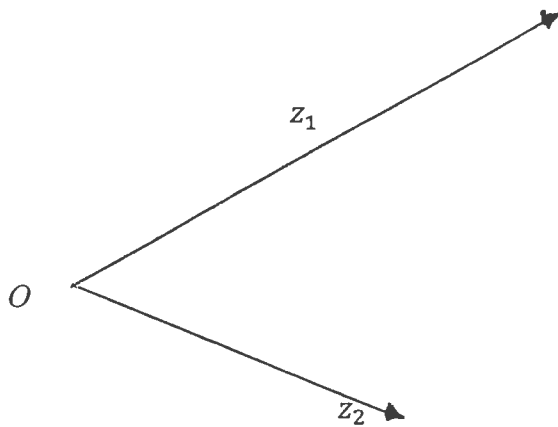


Question 1.**Marks**

- | | | |
|-----|--|---|
| (a) | Find the modulus of $7 + 24i$ | 1 |
| (b) | Expand and simplify $(2 - 3i)(4 + 3i)$ | 2 |
| (c) | Write in the form $a+ib$ ($a, b \in R$): $\frac{3-2i}{1+4i}$ | 2 |
| (d) | Expand and simplify $(z - 1 - 3i)(z + 5 - i)$ | 2 |
| (e) | A square $OABC$ has $\overrightarrow{OA} = 3 + 4i$.
Find the co-ordinates of B and C . | 4 |
| (f) | Describe the transformation when z is multiplied by $-\sqrt{3} - i$ | 2 |
| (g) | Construct the vector $z_1 + z_2$ in the space below.
Clearly show the vector $z_2 - z_1$. | 2 |

**Question 2. (Start A New Page)**

- | | | |
|-----|---|---|
| (a) | Write in the form $a+ib$, where a, b are integers : $(2 - 3i)^3$ | 2 |
| (b) | (i) Shade the region defined by : $3 \leq z + 2 \leq 4$ and $\frac{\pi}{6} \leq \text{Arg}(z + 2) \leq \frac{\pi}{2}$. | 2 |
| | (ii) Find the area of the shaded region. | 2 |
| (c) | Describe the locus of z : $z\bar{z} - 2z - 2\bar{z} - 15 = 0$ | 2 |
| (d) | (i) Find the values of z such that $z^2 = 5 - 12i$ | 2 |
| | (ii) Solve the equation $z^2 + (4i - 7)z + 7 - 11i = 0$ | 2 |
| | (iii) Solve and describe the solutions of :
$ z^2 + (4i - 7)z + 7 - 11i = z - 2 + i $ | 3 |

- (a) Given : $\alpha = 3 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$, $\beta = 2 \left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right)$ and $\gamma = -1 + i$
- (i) Express α^4 in the form $a+ib$ ($a, b \in R$) 2
- (ii) Express $\alpha\beta$ in Modulus / Argument form. 2
- (iii) Express $\alpha\gamma$ in Modulus / Argument form. 2
- (b) (i) Solve over the set of Complex numbers the equation $z^3 = 1$. 2
Write the solutions in the form $a+ib$ where $a, b \in R$.
- (ii) Plot the solutions of $z^3 = 1$ on the Argand Diagram. (Use 1 unit = 2cm) 2
- (iii) Show that the solutions of $z^3 = 1$ can be represented by $1, \omega, \omega^2$, where ω is a non real solution. 2
- (iv) Hence solve in terms of ω the equation : $(1+z)^3 = (1-z)^3$ 3

Question 4. (Start A New Page)

- (a) A triangle ABC is represented by the position vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ by $1+5i$, $3+6i$ and $4-i$ respectively.
- (i) Draw the triangle ABC on the Argand Diagram. (Use 1 unit = 1cm) 1
- (ii) Find the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$. 2
- (iii) Find the Complex number, M , representing the median from C in ΔABC , hence find the length of the median. 2
- (b) A polynomial $P(z)$ is defined by : $P(z) = z^6 + 4z^4 + 16z^2 + 64$.
- (i) Find $P(2i)$ 1
- (ii) Deduce a quadratic factor of $P(z)$ giving reasons. 1
- (iii) Factorise the polynomial $P(z)$ over the set of real numbers R . 3
- (c) Find the Cartesian equation of the locus of: $\left| \frac{z-3}{z+3} \right| = 2$. 3
Describe the locus .
- (d) A function $P(z)$ is always a non real number for all values of z . 2
If $P(z)P(\bar{z}) = 16$ find the value of $|P(z)|$ giving reasons.

End of Exam

*** PLEASE ATTACH THE EXAM PAPER TO QUESTION 1.***

MATHS EXT 2 ASSESSMENT TEST 1
TERM 4, 2012

MATHEMATICS Extension 2: Question 1

Suggested Solutions

(a) $|7+24i| = \sqrt{7^2+24^2} = 25$

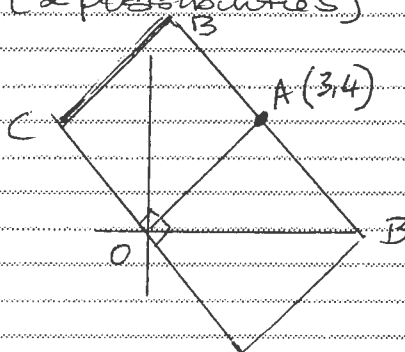
(b) $(2-3i)(4+3i) = (2-3i)(4-3i) = 8-9-18i = -1-18i$

(c) $\frac{3-2i}{1+4i} = \frac{3-2i}{1+4i} \times \frac{1-4i}{1-4i} = \frac{3-12i-2i-8}{1+16} = \frac{-5-14i}{17} = -\frac{5}{17} - \frac{14i}{17}$

(d) $(2-1-3i)(2+5-i) = 2^2 + 5z - i^2z - 2 - 5 + i - 3iz - 15i - 3 = z^2 + 4z - 4iz - 8 - 14i$

(e) Rotation could be anticlockwise or clockwise

(2 possibilities)



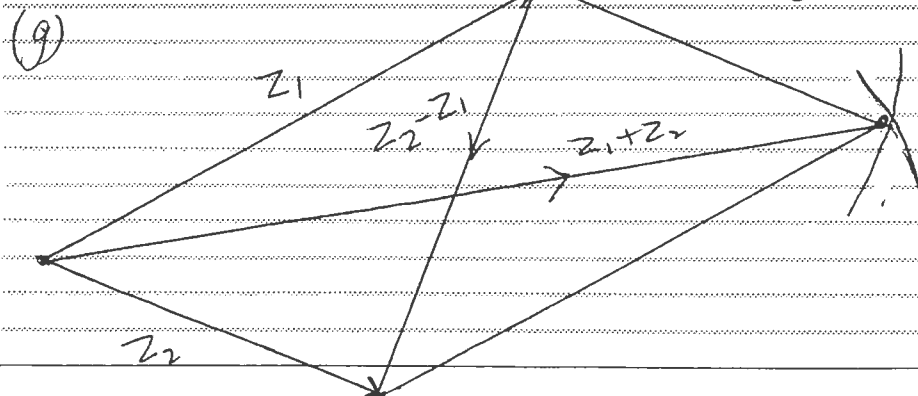
anticlockwise
($\times i$)
 $\vec{OC} = (3+4i) \times i = 3i - 4$
 $\vec{OB} = \vec{OA} + \vec{AB} = \vec{OA} + \vec{OC} = (3+4i) + (-4+3i) = (-1+7i)$

Clockwise ($\times -i$)

$\vec{OC} = (3+4i) \times -i = (-3i + 4)$
 $\vec{OB} = \vec{OA} + \vec{AB} = \vec{OA} + \vec{OC} = (3+4i) + (4-3i) = (7+i)$
Coordinates: C(-4, 3) B(-1, 7) OR C(4, -3) B(7, 1)

(f) $(-\sqrt{3}-i) = 2\text{cis}(-\frac{5\pi}{6})$ or $2\text{cis}(\frac{7\pi}{6})$

$\therefore z$ is enlarged by a factor of 2 and rotated clockwise about 0 by $\frac{5\pi}{6}$ or anticlockwise by $\frac{7\pi}{6}$



Marks

Marker's Comments

①

②

②

②

④

① + ① each term

①

② + ② must write in $a+ib$ form ($\frac{1}{2}$)

1 set of
① C coordinates
② B coordinates

① other set of coordinates

(-1) if coordinates not written

① factor of 2 enlargement

① correct rotation

① must show construction arcs used.

② (z_1+z_2)

② accuracy

② z_1-z_2 (accuracy + direction)

MATHEMATICS Extension 2: Question... 2

Suggested Solutions

Marks

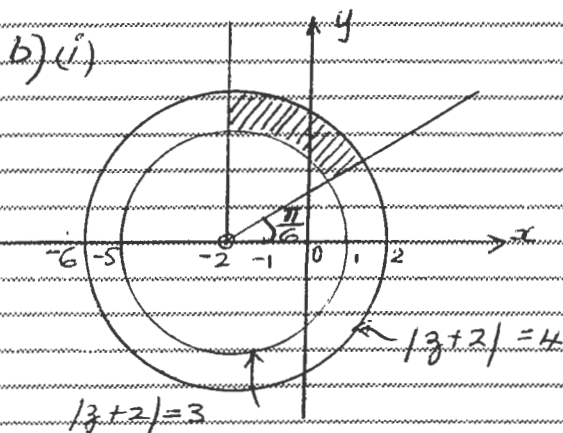
Marker's Comments

$$\begin{aligned} 2a) (2-3i)^3 &= 2^3 + 3(2)^2(-3i) + 3(2)(-3i)^2 + (-3i)^3 \\ &= 8 - 36i - 54 + 27i \\ &= -46 - 9i \end{aligned}$$

2

Students that used the binomial expansion did not understand the concept of coefficients & Pascal's Triangle

$$\begin{aligned} \text{Alternatively, } (2-3i)^3 &= (4-12i-9)(2-3i) \\ &= (-5-12i)(2-3i) \\ &= -10+15i-24i+36i^2 \\ &= -10-9i-36 \\ &= -46-9i \end{aligned}$$



2

This question was done well

$$\begin{aligned} \text{(ii.) Area} &= \frac{1}{2} (R^2 - r^2) \theta \\ &= \frac{1}{2} (4^2 - 3^2) \left(\frac{\pi}{2} - \frac{\pi}{6} \right) \\ &= \frac{7}{2} \times \frac{\pi}{3} \end{aligned}$$

$$\therefore \text{Area} = \frac{7\pi}{6} \text{ units squared}$$

Poorly done
Students confused the Year 8 & 11 Formulae

$$A = \pi r^2 \times \frac{\theta^\circ}{360^\circ}$$

$$A = \frac{1}{2} r^2 \theta^\circ$$

c) Describe the locus $z: z\bar{z} - 2z - 2\bar{z} - 15 = 0$

$$\text{Let } z = x + iy$$

$$\text{then } \bar{z} = x - iy$$

$$z\bar{z} - 2z - 2\bar{z} - 15 = 0$$

$$x^2 + y^2 - 2(2x) - 15 = 0$$

$$x^2 + y^2 - 4x - 15 = 0$$

$$(x-2)^2 + y^2 = 15 + 4$$

$$(x-2)^2 + y^2 = 19$$

The locus is a circle, centre (2,0), radius $\sqrt{19}$ units

MATHEMATICS Extension 2: Question 2...

Suggested Solutions

Marks

Marker's Comments

2 d) i) Find the values of z , such that $z^2 = 5 - 12i$

2

This question was well done apart from careless errors

Let $z = x + iy$, then

$$(x + iy)^2 = 5 - 12i$$

$$x^2 - y^2 + 2xyi = 5 - 12i$$

$$x^2 - y^2 = 5 \quad \text{--- (1) on equating real & imaginary parts}$$

$$2xy = -12 \quad \text{--- (2)}$$

From (2) $y = \frac{-6}{x}$, substitute into (1)

$$x^2 - \frac{36}{x^2} = 5$$

$$x^4 - 5x^2 - 36 = 0$$

$$(x^2 - 9)(x^2 + 4) = 0$$

$$x^2 = 9 \text{ or } x^2 = -4$$

$$\therefore x = \pm 3 \Rightarrow \text{no real solutions}$$

$$\therefore y = \mp 2$$

$$\therefore z = \pm (3 - 2i)$$

ii) Solve $z^2 + (4i - 7)z + 7 - 11i = 0$

2

This question was well done apart from careless errors.

$$z = \frac{7 - 4i \pm \sqrt{(4i - 7)^2 - 4(7 - 11i)}}{2}$$

$$= \frac{7 - 4i \pm \sqrt{-16 + 49 - 56i - 28 + 44i}}{2}$$

$$= \frac{7 - 4i \pm \sqrt{5 - 12i}}{2}$$

$$= \frac{7 - 4i \pm (3 - 2i)}{2} \quad \text{from (i)}$$

$$= \frac{10 - 6i}{2} \text{ or } \frac{4 - 2i}{2}$$

$$= 5 - 3i \text{ or } 2 - i$$

(iii) $|z^2 + (4i - 7)z + 7 - 11i| = |z - 2 + i|$

3

$$|(z - (5 - 3i))(z - (2 - i))| = |z - (2 - i)|$$

$$|z - 5 + 3i| |z - 2 + i| = |z - 2 + i|$$

$$|z - 5 + 3i| = 1$$

$\therefore z = 2 - i$ The point $(2, -1)$

OR $|z - (5 - 3i)| = 1$, the circle $(5, -3)$, radius 1 unit.

Students got to this line but then claimed that if $|x||y| = |p|$ then $|x| = |p|$ or $|y| = |p|$ which is incorrect assumption.

Suggested Solutions

Marks

Marker's Comments

$$a: i) \alpha^4 = \left(3 \operatorname{cis} \frac{\pi}{6}\right)^4 = 81 \operatorname{cis} \frac{2\pi}{3}$$

/

$$= 81 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$= -\frac{81}{2} + i \frac{81\sqrt{3}}{2} \#$$

/

Some get the sign wrong -1m

$$ii) \alpha \beta = 3 \operatorname{cis} \frac{\pi}{6} \cdot 2 \operatorname{cis} \left(-\frac{\pi}{4}\right)$$

/

$$= 6 \operatorname{cis} \left(\frac{\pi}{6} - \frac{\pi}{4}\right)$$

$$= 6 \operatorname{cis} \left(-\frac{\pi}{12}\right)$$

$$= 6 \left(\cos -\frac{\pi}{12} + i \sin -\frac{\pi}{12} \right) \#$$

/

Need to write out the expanded form -1/2m

$$iii) \alpha \gamma = 3 \operatorname{cis} \frac{\pi}{6} \cdot \sqrt{2} \operatorname{cis} \frac{3\pi}{4}$$

/

$$= 3\sqrt{2} \operatorname{cis} \left(\frac{\pi}{6} + \frac{3\pi}{4}\right)$$

$$= 3\sqrt{2} \operatorname{cis} \left(\frac{11\pi}{12}\right) = 3\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)$$

/

Many students write $\delta = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$ max 1mSome do not know Mod-Arg. FORM
or $6 \left(\cos \frac{\pi}{12} - i \sin \frac{\pi}{12} \right)$ is not.

$$b: i) \text{ Let } z = r \operatorname{cis} \theta$$

$$z^3 = r^3 \operatorname{cis} 3\theta = 1$$

$$r^3 \operatorname{cis} 3\theta = 1^3 \operatorname{cis} (0 + 2n\pi)$$

$$n \in \mathbb{Z}$$

$$\text{or } n = 0, 1, 2$$

$$\therefore r = 1, \quad \theta = \frac{2n\pi}{3}$$

/

Generally well done

$$n=0$$

$$z = \operatorname{cis} 0 = 1 + 0i$$

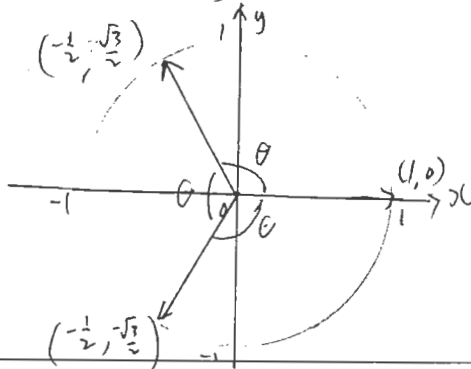
$$n=1$$

$$\omega = \operatorname{cis} \frac{2\pi}{3} = \frac{1}{2} [-1 + i\sqrt{3}]$$

$$n=2$$

$$\omega^2 = \operatorname{cis} \frac{4\pi}{3} = \frac{1}{2} [-1 - i\sqrt{3}]$$

ii)



$$\theta = \frac{2\pi}{3}$$

2

Need to answer in $a+bi$ form otherwise max. 1m $\frac{1}{2}$ m measuring & label angle $\frac{1}{2}$ m radius (2cm) + intercepts1 m for circle with labelled z_1, z_2, z_3

7/11 2012 T4

MATHEMATICS Extension 1: Question 3

P.2/2

Suggested Solutions

Marks

Marker's Comments

(iii) From (ii)

$$z = 1, \quad \text{cis } \frac{2\pi}{3} \text{ or } \text{cis } \frac{4\pi}{3}$$

$$\text{Let } w = \text{cis } \frac{2\pi}{3}$$

$$w^2 = \left(\text{cis } \frac{2\pi}{3}\right)^2 = \text{cis } \left(\frac{2\pi}{3} \times 2\right) = \text{cis } \frac{4\pi}{3}$$

$\therefore 1, w, w^2$ are roots of $z^3 = 1$

Alternatively let $w = \frac{1}{2}[-1 + i\sqrt{3}]$

$$w^2 = \left[\frac{1}{2}(-1 + i\sqrt{3})\right]^2 = \frac{1}{4}(1 - 3 - 2i\sqrt{3})$$

$$w^2 = \frac{1}{4}(-2 - 2i\sqrt{3}) = \frac{1}{2}(-1 - i\sqrt{3})$$

$\therefore 1, w, w^2$ are roots of $z^3 = 1$

or $z^3 = 1$ and w is a non real root

$$\therefore w^3 = 1$$

$$(w^3)^2 = (w^2)^3 = 1$$

and w, w^2 are different in part i

$\therefore w^2$ is also a root

$$(iv) (1+z)^3 = (1-z)^3$$

$$\left(\frac{1+z}{1-z}\right)^3 = 1 \quad z \neq 1$$

$$\text{From (i)} \quad \frac{1+z}{1-z} = 1, \quad w \text{ or } w^2$$

$$\frac{1+z}{1-z} = w^n \quad n = 0, 1, 2$$

$$1+z = w^n - zw^n$$

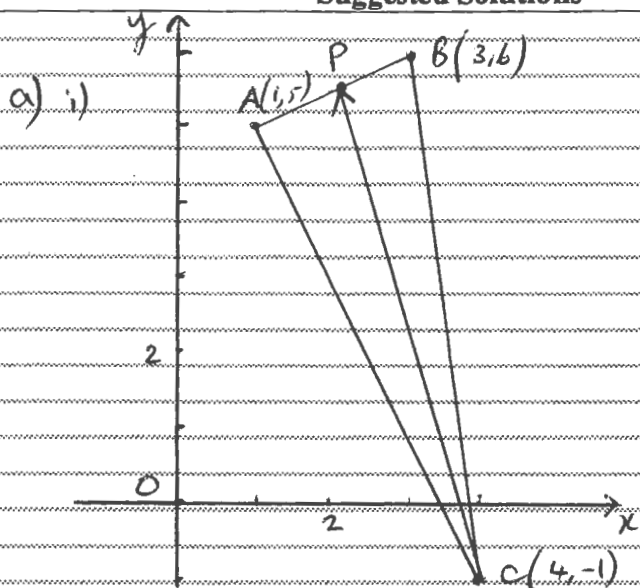
$$z = \frac{w^n - 1}{w^n + 1}$$

$$z = 0, \quad \frac{w-1}{w+1} \text{ or } \frac{w^2-1}{w^2+1}$$

Suggested Solutions

Marks

Marker's Comments



ii) $\overrightarrow{AB} (= \overrightarrow{OB} - \overrightarrow{OA})$ represents $(3-1) + i(6-5)$
 $= \underline{\underline{2 + i}}$

$\overrightarrow{AC} (= \overrightarrow{OC} - \overrightarrow{OA})$ represents $(4-1) + i(-1-5)$
 $= \underline{\underline{3 - 6i}}$

iii) The vector $\overrightarrow{CP}$ as shown represents the complex number M .

P is the point $(2, 5\frac{1}{2})$

$\therefore \overrightarrow{CP} (= \overrightarrow{OP} - \overrightarrow{OC})$ represents $(2-4) + i(5\frac{1}{2}-(-1))$
 $\therefore M = \underline{\underline{-2 + \frac{13i}{2}}}$

Hence median length $= |M|$

$= \sqrt{4 + \frac{169}{4}} = \underline{\underline{\frac{\sqrt{185}}{2}}}$ units

b)

i) $P(2i) = -64 + 4(16) + 16(-4) + 64$
 $= \underline{\underline{0}}$

ii) $(z - 2i)$ is a factor by Remainder/Factor theorem. Since coefficients are real, complex roots come in conjugate pairs

Diagrams generally fine (surprising what guidance with scales will do)

Occasional issue with signs but generally fine

Many people mixed up M with the midpoint of AB .

Although they got the length right (with correct working)

Suggested Solutions	Marks	Marker's Comments
$\therefore (z+2i)$ is also a factor. $\therefore (z-2i)(z+2i) = \underline{(z^2+4)}$ is a real quadratic factor. iii) $P(z) = (z^2+4)(z^4+16)$ by long division or inspection. <u>Method 1</u> $(z^4+16) = (z^4+8z^2+16) - 8z^2$ $= (z^2+4)^2 - (2\sqrt{2}z)^2$ $= (z^2+4-2\sqrt{2}z)(z^2+4+2\sqrt{2}z)$ $\therefore P(z) = \underline{(z^2+4)(z^2-2\sqrt{2}z+4)(z^2+2\sqrt{2}z+4)}$ <u>Method 2</u> Solve $z^4 = -16$ $= 2^4(\text{cis } \pi + 2n\pi)$ $n \in \mathbb{Z}$ $\therefore z = 2 \text{cis} \left(\frac{\pi + n\pi}{4} \right)$ (de Moivre) $= 2 \text{cis} \left(-\frac{3\pi}{4} \right), 2 \text{cis} \left(-\frac{\pi}{4} \right), 2 \text{cis} \left(\frac{\pi}{4} \right), 2 \text{cis} \left(\frac{3\pi}{4} \right)$ Combine pairs of linear factors $(z - 2 \text{cis} \frac{\pi}{4})(z - 2 \text{cis} (-\frac{\pi}{4})) = z^2 - 4 \cos \frac{\pi}{4} z + 4$ $= z^2 - 2\sqrt{2}z + 4$ Similarly $(z - 2 \text{cis} \frac{3\pi}{4})(z - 2 \text{cis} (-\frac{3\pi}{4})) = z^2 + 2\sqrt{2}z + 4$ $\therefore P(z) = \underline{(z^2+4)(z^2-2\sqrt{2}z+4)(z^2+2\sqrt{2}z+4)}$ c) Let $z = x+iy$ $x, y \in \mathbb{R}$ $z-3 = 2 z+3$ ($z \neq 3$) $(x-3)+iy = 2 (x+3)+iy$ $\therefore (x-3)^2 + y^2 = 4\{(x+3)^2 + y^2\}$	1 $\frac{1}{2}$ $2\frac{1}{2}$ $2\frac{1}{2}$	$\frac{1}{2}$ mark for reason Only $\frac{1}{2}$ mark for this which is where many people stopped! Method 1 is clearly quicker but method 2 is standard book-work and should have been accessible to all.

Suggested Solutions	Marks	Marker's Comments
c) (continued) $x^2 - 6x + 9 + y^2 = 4x^2 + 24x + 36 + 4y^2$ $3x^2 + 3y^2 + 30x + 27 = 0$ $x^2 + 10x + y^2 = -9$ $(x + 5)^2 + y^2 = 16$ <u>This is a circle, centre $(-5, 0)$, radius 4</u>	2 1	Most people got this. Some small errors. (1/2 off if add/sub) (1 off if $2^2 = 2$!!)
d) (Original question invalid)		